

Failure Mode and Effects Analysis (FMEA)

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Product:	NEXUS CK MB (30421)
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1. Purpose

This FMEA evaluates potential failure modes in the manufacturing process of NEXUS CK MB (Reference 30421) immunoassay kits. The analysis covers all critical process steps from sensor cartridge receipt through final packaging, with particular attention to calibration-related risks.

2. Scope

Applicable to the complete NEXUS CK MB manufacturing process at the Riverside production facility. Process steps: CART Receipt, card Assembly, Kit Assembly, Calibration, QC Testing, Labeling, Packaging.

3. FMEA Risk Assessment

Process Step	Failure Mode		Effect	S	Cause	O		
Controls	D	RPN						
CART_Receipt Incoming inspection per SOP-CART-001	4	96	CART coating variation	Signal out of specification	8	Supplier lot-to-lot variation	3	
Calibration Quarterly PM, trend monitoring	5	90	Calibration drift	Invalid calibration (B18)	9	Equipment aging/wear	2	
Kit_Assembly inspection, barcode scan	3	27	Wrong component	Incorrect test results	9	Labeling error	1	Visual
Strip_Assembly nozzle verification, flow check	3	42	Reagent deposition error	Reduced assay sensitivity	7	Dispenser nozzle clog	2	Daily

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Process Step	D	Failure Mode RPN	Effect	S	Cause	O	Controls
Qualification programme, double-check QC_Testing		Incorrect sample handling 3 36	False QC result	6	Operator training gap	2	
procedure							
Labeling verification at labeling station	2	Mislabeled kit 14	Product traceability loss	7	Printer malfunction	1	Barcode
Packaging check, visual inspection checklist	3	Incomplete kit content 36	Customer complaint	6	Missing component during assembly	2	Weight
CART_Receipt ERP lot tracking, FIFO storage labels		Expired sensor cartridges used 2 16	Degraded antibody performance	8	FIFO not followed in storage		1

4. Risk Priority Summary
Critical threshold: RPN >= 80. Two failure modes exceed the critical threshold and require enhanced monitoring:
- CART coating variation (RPN=96): Incoming inspection enhanced with antibody density measurement per SOP-CART-001.
- Calibration drift (RPN=90): Quarterly preventive maintenance and continuous trend monitoring of S1/S2 signals.

5. Revision History				
Version	Date	Author	Changes	
1.0	2021-06-01	J. Martin	Initial release	
1.5	2022-03-15	J. Martin	Updated RPN for calibration drift after INV-2022-0045	
2.0	2023-01-15	J. Martin	Added CART coating variation risk; updated controls	

